

## Study Questions: Cognition

1. What does Active Inference mean by "beliefs are probability distributions"? How does this differ from the folk-psychological concept of belief?
2. Compare Fodor's "language of thought" hypothesis with the Active Inference account of belief. What are the key differences?
3. In what sense are beliefs "physical states" in Active Inference? Does this commit the framework to physicalism?
4. Explain the role of precision in Active Inference. How does it relate to the philosophical concept of attention?
5. How does Active Inference account for selective attention? Give an example.
6. What is the "dark room problem"? Why is it considered a challenge to Active Inference?
7. How do prior preferences (the C vector) resolve the dark room problem? What philosophical assumptions does this resolution require?
8. Compare the Active Inference resolution of the dark room problem with Aristotle's concept of *eudaimonia* (flourishing).
9. How might aberrant precision underlie anxiety? Sketch the Active Inference account and compare it with a cognitive-behavioral account.
10. What is the "hard problem of consciousness" (Chalmers), and does Active Inference address it?
11. Can inference occur without consciousness? What are the philosophical implications of unconscious inference?
12. How does the phenomenology of "flow states" (Csikszentmihalyi) relate to precision dynamics in Active Inference?
13. If beliefs are updated continuously, is there a stable "self" that holds beliefs? Or is the self itself a belief?
14. How does the concept of precision relate to the philosophical notion of certainty (Descartes, Wittgenstein)?
15. What is the relationship between cognition and metacognition in Active Inference? Can an agent have beliefs about its own beliefs?
16. How does Active Inference handle the distinction between intuitive and reflective cognition (Kahneman's System 1 vs System 2)?

17. Evaluate the claim: "Cognition is just perception at a higher level of abstraction."
18. How does the extended cognition hypothesis interact with the Active Inference account of beliefs as physical states?
19. Can Active Inference explain creativity? If cognition is inference, where does novelty come from?
20. Construct an argument for or against the claim: "Active Inference shows that the mind-body problem is dissolved, not solved."